Feedback Is a Hypothesis: Benchmarking Evidence-Scoped Revision of Multi-Claim Language Model Responses

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Abstract

When a user corrects a language model, the correction may be right, partly right, vague, outdated, or wrong, and a multi-claim response contains claims the correction touches directly, claims that were inferred from them, and claims it should leave alone. Existing benchmarks split this problem: edit-propagation benchmarks label direct and downstream updates but assume the requested edit is correct, while sycophancy benchmarks test whether models resist false feedback but score only whole answers. We introduce **IntExR-Scope**, a claim-level revision task in which feedback is an unverified hypothesis and evidence is the arbiter. Given an original response, prior evidence, feedback of unknown validity, and new evidence, a system must label every claim as a direct update, a propagated update, unresolved (to be hedged, not deleted), or keep, judge the feedback, and write a minimal revision. We release a validated benchmark built from clinical notes with expert-injected errors (MedScope: 40 source cases, 248 examples, 1,186 claims across eight feedback conditions, with dependency chains up to depth two) together with converted non-medical sets, an evaluation harness with case-level uncertainty, and a *scope-first* method that delegates evidence alignment and wording to an LLM but assigns labels and preserves untouched claims deterministically. Our central experiment compares the same claims under a verified edit and under unverified feedback plus evidence, isolating the cost of verification on scope and propagation. **[TODO:** one sentence of headline findings once runs are in.]

1 Introduction

A clinician reads an AI-drafted assessment and plan and says, "This doesn't look like CJD to me." Whether the system should change its diagnosis depends on more than the remark: an MRI showing frontotemporal atrophy and a negative RT-QuIC settle it, an attending's say-so without either does not, and once the diagnosis changes, the prognosis and the tests that were ordered *because of* it must change too, while the history, the exam findings, and the caregiver plan must stay exactly as written. Getting this right requires three judgments that current evaluations treat separately: *whether* the feedback is right (it is a hypothesis to check), *which* claims the evidence changes directly, and *which* claims change only because something they depend on changed.

Two recent benchmarks make the scope part concrete. EditPropBench (Kruthof, 2026) and RevPropBench (Kikuta, 2026) label direct targets, required downstream updates, and protected content in documents and structured artifacts, and find that strong models miss a large share of downstream updates. Both, however, hand the model an edit that is correct by construction. In deployed systems the edit is a user's claim. Sycophancy research (Sharma et al., 2024; Fanous et al., 2025; Stengel-Eskin et al., 2025) shows that models often accept false corrections and sometimes reject true ones, but it scores one decision per interaction: a whole answer flips or it does not. Neither line can express the most common clinical case, in which feedback is right about one claim, silent about its consequences, and wrong or unsupported about something else.

We argue that the two problems must be studied jointly, because verification changes scope. If a model must first decide whether a correction is warranted, the set of claims it revises is no longer given; a wrong correction about a claim that the new evidence overturns anyway still requires a change, a correct correction with no supporting evidence calls for a hedge rather than a flip, and a vague complaint must not license edits that no evidence supports. The principle we build on is simple: **feedback is a hypothesis; evidence is the arbiter**.

Contributions.

1. **Task.** IntExR-Scope (§3): a claim-level revision task with four labels (Direct, Propagated, Unresolved, Keep), a precedence rule that makes the labels decidable, an unresolved label that itself propagates, and four input conditions that separate the cost of verifying feedback from the cost of propagating it.
2. **Benchmark.** MedScope (§4), built on real clinical vignettes with expert-injected errors (Ben Abacha et al., 2025): each source case yields up to eight feedback conditions (valid, false, partial, authority-only, vague, outdated, superseding, conflicting evidence) over identical claims, so that feedback validity is varied while the response is held fixed. We also convert three public datasets into the same format.
3. **Method.** A scope-first pipeline (§5) in which an LLM aligns claims to evidence and words the revisions, while label assignment, preservation of kept claims, and a feedback-leak check are deterministic. With perfect alignment it reproduces every gold label in MedScope, which makes the label rules executable.
4. **Evaluation.** An open, dependency-free harness with change recall split by direct and propagated claims and by dependency depth, preservation error on evidence-supported claims, unresolved-claim F1, and case-level bootstrap intervals; and a study of **[TODO: N]** systems across **[TODO: M]** open and API models under all four conditions (§7).

2 Related Work

Edit propagation. Knowledge-editing evaluations test whether a parametric edit reaches its logical consequences (Zhong et al., 2023; Cohen et al., 2024; Onoe et al., 2023). EditPropBench (Kruthof, 2026) moves this to text: synthetic manuscripts with fact graphs, labelled direct targets, required downstream updates, and protected text; on its implicit stratum the best system misses about 30% of required updates and a plan-then-edit variant underperforms direct editing. RevPropBench (Kikuta, 2026) labels direct versus cascade edits in JSON artifacts across nine domains and explicitly excludes ambiguous cases. LEDGER (LEDGER authors, 2026) edits documents through an explicit dependency graph. All three assume that the requested change is correct and none allows a claim to be left unresolved; our condition A reproduces their setting so that the difference can be measured on the same claims.

Evidence-based revision. FRUIT (Logan IV et al., 2022) updates Wikipedia articles from new evidence, RARR (Gao et al., 2023) revises unattributed spans while measuring preservation, and claim-level correction methods decompose, verify, and edit (Thorne and Vlachos, 2021; Dhuliawala et al., 2024; Mishra et al., 2024). They treat evidence as the oracle and rarely label the claims that depend on a fix. Clinical correction benchmarks (Ben Abacha et al., 2025; Zou et al., 2025) correct one injected error per note or report and do not measure whether correct content survives.

Sycophancy and persuasion. Models capitulate to user pushback (Sharma et al., 2024; Laban et al., 2023), including in medicine (Fanous et al., 2025; Xiao et al., 2026). Persuasion-balanced training (Stengel-Eskin et al., 2025) shows that resisting bad persuasion and accepting good persuasion trade off. These studies score whole answers; IntExR-Scope scores the partition of a response, so a reply that rightly corrects one claim but also deletes a supported one is penalised for the second.

Belief revision and abstention. AGM revision (Alchourrón et al., 1985) formalises minimal change, Katsuno and Mendelzon (1991) separate revising beliefs about a fixed world from updating them after the world changes (our *superseding* condition), and truth-maintenance systems retract beliefs whose justifications are withdrawn (Doyle, 1979; de Kleer, 1986), the formal ancestor of our P label. LLM belief-revision benchmarks use synthetic logic (Wilie et al., 2024). Abstention work studies answer-level refusal (Kirichenko et al., 2025); our U label asks for claim-level hedging inside a revision.

3 Task: IntExR-Scope

Input. An example consists of a context x (the original request), an original response R decomposed into ordered claims $C = \{c_1, \ldots, c_n\}$ that are spans of R, prior evidence E_0, feedback F from a speaker with a role, and new evidence E_1. Every evidence item has an identifier and a provenance. In this paper the decomposition is given, which isolates scope prediction from claim extraction; end-to-end decomposition is left to future work. The dependency graph G ($c_j \to c_i$ if c_i was

inferred from c_j), the labels, and the validity of F are hidden.

Labels. Each claim receives exactly one label.

- **D (direct update):** some evidence item, read on its own, contradicts c, refines it in a way that changes what follows from it, or makes a forward-looking claim moot. Feedback alone never triggers D.
- **P (propagated update):** no evidence item addresses c, but an ancestor in G is D or P and c no longer holds once the ancestor is revised.
- **U (unresolved):** the evidence about c conflicts or is inconclusive (U-evidence); c depends on a U claim (U-dependency); or feedback disputes c and no evidence decides it (U-feedback). The target behaviour is to keep the claim but state the uncertainty and what would resolve it.
- **K (keep):** none of the above. The claim must remain semantically unchanged. We record whether evidence actively supports it (K^+).

When several labels could apply, **direct evidence** > **dependency** > **unsupported feedback** > **keep**; within direct evidence, newer items override older ones. The rule makes labels decidable from evidence and G and is what the deterministic stage of our method implements (§5).

Feedback verdict. Systems also judge the feedback as a whole: *accept* (everything it asserts is supported), *partial*, *reject* (contradicted, or about something the response does not describe), or *insufficient* (too vague, or undecided by the evidence). The verdict is independent of the claim labels: feedback can be rejected while the evidence still forces changes.

Conditions. To separate verification from propagation we evaluate the same claims under four inputs: **(A)** verified edit: F is replaced by the gold direct edits stated as correct (the setting of EditPropBench and RevPropBench); **(B)** unverified feedback plus evidence (the default); **(C)** evidence only; **(D)** feedback only, which measures pure deference to feedback. The quantity of interest is the paired difference B−A on identical claims, the *cost of verification*.

Metrics. Free-text systems are mapped to claim actions (*keep, revised, removed, hedged*) by a claim-alignment judge, which we calibrate against human labels (§6); structured systems report actions directly and are additionally checked against the judge. We report **change recall** on gold D and on gold P claims (the headline for "is propagation harder?"), split by dependency depth; **preservation error**, the fraction of K (and K^+) claims changed, removed, or hedged; **U-F1**, treating hedging as predicting U; **verdict accuracy**; the rate of revisions free of **forbidden additions** (content asserted only by feedback); deterministic **text checks** on revised claims; and **interaction-exact** accuracy, all claims of an example handled correctly. Structured systems additionally get direct-scope and propagation F1. Confidence intervals are 95% percentile bootstraps over *source cases*, never over claims or feedback variants of the same case.

4 The MedScope Benchmark

4.1 Construction

MedScope starts from MEDEC (Ben Abacha et al., 2025), clinical vignettes in which medical experts injected one error (diagnosis, management, pharmacotherapy, or treatment) and wrote its correction (CC BY 4.0). For each selected note we remove the error sentence and use the remainder as prior evidence E_0. The original response R is an assessment and plan that *contains the error and claims inferred from it*, for example a wrong diagnosis followed by the treatment and monitoring that follow from that diagnosis, plus claims unaffected by it. New evidence E_1 is a decisive result or guideline statement. Every source case is instantiated under the following conditions, which keep R and C fixed and vary only feedback and evidence:

- **valid**: correct feedback about the error, decisive evidence ⇒ authored D/P, accept.
- **false**: feedback attacks a correct claim; an added evidence item supports that claim ⇒ it stays K^+, the evidence still forces the D/P updates, reject.
- **partial**: right about the error but adds an unsupported intervention, which becomes a forbidden addition, partial.
- **authority-only**: the valid feedback from a department chief with *no* new evidence. If the note itself decides the error the labels stand;

otherwise D becomes U-feedback and its dependents U-dependency, insufficient.

- **ambiguous**: generic vague feedback ("something doesn't sit right"); evidence still decides the claims; insufficient.
- **outdated**: feedback defends the error by citing an older note; the newer evidence wins; reject.
- **superseding** (5 cases): an event *after* the plan was written changes a claim that was correct when written, the revision-versus-update distinction of Katsuno and Mendelzon (1991).
- **conflicting** (3 cases): a second read contradicts the decisive evidence ⇒ U-evidence for the erroneous claim and U-dependency below it.

Quality control. A validator rejects any example in which a claim is not a span of R, a P claim lacks a changed parent, a U-dependency lacks a U parent, a deleted claim is U, or an evidence identifier is unknown. The builder asserts that every forbidden addition is absent from R, that every text check is satisfied by the gold revision and violated by R, and that false feedback targets a K claim. The gold oracle scores 1.0 on every metric, and our method's deterministic stages reproduce all gold labels from gold alignments (§5).

Review status. The authored parts of MedScope (responses, new evidence, feedback, target texts) were proposed with LLM assistance and have not yet been reviewed by clinicians. [**TODO:** Clinician review protocol and results; second annotator and Cohen's κ on a double-labelled subset; adjudication. Until then all MedScope results are reported as pilot results.]

4.2 Statistics

Table 1 summarises the data. MedScope averages 4.8 claims per example; 26% of its claims are P, more than its D claims, because each error has several dependents, and 59 P claims sit at depth two (error → treatment → monitoring).

4.3 Converted public sets

To test the claims outside medicine and against the closest prior benchmark, we convert three datasets into the IntExR format with valid and false feedback variants. **EditPropBench-IX**: synthetic manuscripts from EditPropBench (Kruthof, 2026); the evidence states only the explicit edited facts, units stating them are D, and required units that carry only an implied qualifier are P, so the set tests propagation under verification. **VitaminC-IX**: contrastive Wikipedia revisions (Schuster et al., 2021) whose not-enough-info cases provide U-evidence. **MEDEC-knowledge**: MEDEC notes whose error can be found only from medical knowledge, with no evidence item; it is a control for evidence-gated methods. [**TODO:** Decide on a second real non-medical source (FRUIT or ARIES); §8.]

	cases	ex.	claims	D	P	K	U
MedScope	40	248	1186	220	309	577	80
P depth 1 / 2					250/59		
K^+ / U by trigger†						40	3/47/30
Pilot (CXR, COCO)	16	16	86	19	16	44	7
EditPropBench-IX	20	40	1600	205	103	1292	0
VitaminC-IX	21	40	160	45	0	86	29
MEDEC-knowledge	25	40	370	30	0	340	0

Table 1: Benchmark statistics. †U triggers: evidence / dependency / feedback. MedScope conditions: 40 each of valid, false, partial, authority-only, ambiguous, outdated; 5 superseding; 3 conflicting. The pilot set is synthetic MIMIC-style chest X-ray reports and real COCO captions; a 24-example version with real IU X-ray reports exists but is not redistributed (CC BY-NC-ND).

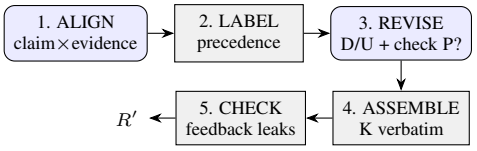


Figure 1: Scope-first pipeline. Shaded boxes are deterministic; blue boxes are LLM calls. CHECK sends claims that contain feedback-only content back to REVISE once, so an example costs two calls, or three with a re-revision.

5 Scope-First Revision

Single-prompt revision asks one model call to verify feedback, find dependents, and rewrite, and leaves every kept claim exposed to rewording. Our method (Figure 1) gives the LLM only the steps that need language understanding.

1. **Align (LLM).** For every claim, list the evidence items that *support*, *contradict*, or leave *uncertain* the claim itself, and the claims it was inferred from; list the claims the feedback disputes and give a verdict. The prompt states that feedback is not evidence, and that a claim is not contradicted merely because a claim it depends on is wrong.

2. **Label (code).** Apply the precedence rule of §3 in topological order of the predicted dependencies: the newest evidence tier that speaks to a claim decides it (D if it contradicts, U if it is uncertain or internally split, K^+ if it supports); otherwise a changed parent makes the claim a P *candidate*, a U parent makes it U, a dispute without evidence makes it U, and anything else is K. Under condition A the verified edits are added as an evidence item.

3. **Revise (LLM).** One call writes new text only for D claims (update or delete), U claims (hedge, with the reason supplied by step 2), and P candidates, for which the model either returns UNCHANGED (the claim still holds and becomes K) or rewrites or deletes it (it becomes P). Kept claims are shown as fixed context.

4. **Assemble (code).** Revised claim texts are spliced into R at their spans; K claims are never regenerated, so preservation of claims labelled K holds by construction and formatting outside the claims is untouched.

5. **Check (code).** Content words that occur in the feedback but in no evidence item, the context, or R are *feedback-only*. Any revised claim containing one is re-revised once with an explicit instruction not to include it.

Executable specification. Because labels are assigned by code, the label rules can be tested: feeding the pipeline gold alignments and gold revision texts reproduces every gold label, verdict, and text check on all 248 MedScope examples, the pilot, and EditPropBench-IX (a unit test in the release). Errors of the full system are therefore attributable to the two LLM stages, which we analyse separately (§7.1).

Ablations. *no-prop* keeps every P candidate (no propagation check); *no-check* removes step 5; and the single-prompt *scope-first* baseline asks one call for the same JSON (labels, dependencies, revised claims, verdict, full revision) without deterministic stages.

6 Experimental Setup

Systems. Following the thesis plan, we compare: **no revision**; **always accept** (a lexical heuristic that marks claims sharing content words with the feedback); **always rewrite** (rewrite the response from all inputs); **dialogue** (the feedback and new evidence pasted as the next chat turn); **evidence-aware prompting** (instructions that feedback may be wrong, to change only what evidence requires including inferred statements, to keep other sentences verbatim, and to hedge undecided claims, which is the "preservation clause" baseline recommended by minimal-edit work); **single-prompt scope-first**; a claim-level **evidence gate** (an NLI model scores each claim–evidence pair; contradiction ⇒ remove, dispute without entailment ⇒ hedge, no propagation); and the **scope-first pipeline** with its ablations. **[TODO:** Strong current correction baseline: LEDGER-style dependency-graph editing if code is available, else RARR-style research-and-revise.**]**

Models. All experiments run at zero monetary cost on public data. Open-weight models are served with vLLM on a university GPU server: **[TODO:** Qwen3-8B, Qwen3-32B (AWQ), Llama-3.3-70B (AWQ)**]**; API models use free tiers: **[TODO:** Gemini Flash; gpt-oss-120b via Groq**]**. Temperature is 0; every response is cached and released. The NLI gate uses **[TODO:** DeBERTa-v3-large NLI checkpoint, id verified on the hub**]**.

Judge and its calibration. Free-text revisions are mapped to claim actions by an LLM judge that sees the original claims and the revision but not the evidence or gold. **[TODO:** Report Cohen's κ between the judge and the first author on 100 stratified (claim, revision) pairs (scripts/label_actions.py); target $\kappa \geq 0.75$; report the confusion matrix and, if below target, switch judge model or add human labels for the affected strata.**]**

Pre-registered hypotheses. We state the predictions before running models. **H1 (cost of verification):** on the same claims, condition B lowers P change recall and interaction-exact accuracy relative to A. **H2 (propagation is harder):** change recall on P is below that on D for every system and falls from depth one to depth two. **H3 (sycophancy at claim level):** in the false condition, free-text systems change evidence-supported K^+ claims at a higher rate than in the valid condition. **H4 (under-hedging):** U-F1 is low for all LLM systems; with authority-only feedback, models flip undecided claims instead of hedging them. **H5 (method):** the scope-first pipeline lowers preservation error and forbidden additions relative to evidence-aware prompting without lowering D change recall; its effect on P recall is an open question, given that

plan-then-edit hurt propagation on EditPropBench.

7 Results

Offline baselines. Table 2 gives final numbers for the systems that need no model. Two observations already hold. First, the lexical always-accept heuristic changes 72% of the evidence-supported claims that false feedback attacks [57–85], so the false condition does measure claim-level deference. Second, its change recall drops from 0.40 on D to 0.17 on P and to 0.10 at depth two: dependents rarely share words with the feedback, so following the feedback's surface finds direct targets but not their consequences. Its D recall is inflated in the outdated condition (1.00), whose feedback quotes the erroneous sentence verbatim; we report this as a known shortcut.

H1–H5. **[TODO:** Fill from results/RESULTS_medscope.md and results/RESULTS_epb.md: (i) the B−A table with paired intervals; (ii) change recall by depth; (iii) per-condition preservation error on K^+; (iv) the authority-only U analysis; (v) pipeline vs. evidence-aware prompt. Report null results as such.]

7.1 Analysis

[TODO: Error audit of 50 pipeline failures: align errors (missed contradiction, false dependency, dispute treated as evidence) vs. revise errors (wrong UNCHANGED, hedge written as flip); judged-from-text vs. self-reported labels for structured systems; cost (tokens, calls) and latency per example.]

8 Limitations

Unreviewed clinical content. The authored parts of MedScope have not yet been reviewed by clinicians, and labels come from one annotator; we will report agreement and adjudication before release, and some cases may be revised or dropped. **Gold claims.** Claims are given; end-to-end decomposition will lower all scores and is not measured. **Text checks** are lexical proxies for "the revision says the right thing"; the calibrated judge is the primary measure. **Knowledge versus evidence.** Our labels require evidence; clinicians also correct from knowledge, which MEDEC-knowledge only probes. **Scale.** 40 source cases give wide case-level intervals; we report them rather than claim-level intervals that would overstate precision. **Templates.** Ambiguous and outdated feedback are templated across cases, which a model could learn; held-out wording is **[TODO:** planned]. **Domain.** The second real non-medical source is undecided (FRUIT or ARIES).

Ethics Statement

All data are public and de-identified: MEDEC (CC BY 4.0), COCO captions (CC BY 4.0), VitaminC (CC BY-SA 3.0), EditPropBench (CC BY 4.0). The pilot chest X-ray reports are synthetic. IU X-ray reports (CC BY-NC-ND 4.0) are used only through identifiers and scripts, and no edited text is redistributed. No credentialed data (e.g., MIMIC) were sent to any external API. The benchmark is for evaluating revision behaviour and is not a source of clinical advice.

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System (condition B)	Chg-D↑	Chg-P↑	PresErr-K↓	PresErr-K$^+$ ↓	U-F1↑	Verdict↑	NoForbid↑	Exact↑
Oracle	1.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00
No revision	0.00	0.00	0.00	0.00	0.00	0.32	1.00	0.00
Always accept (lexical)	0.40	0.17	0.23	0.72	0.00	0.22	—	0.05
Dialogue	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	—	*pend.*	*pend.*
Always rewrite	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	—	*pend.*	*pend.*
Evidence-aware prompt	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	—	*pend.*	*pend.*
Evidence gate (NLI)	*pend.*	0.00*	*pend.*	*pend.*	*pend.*	*pend.*	1.00*	*pend.*
Scope-first (1 prompt)	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*
Scope-first pipeline	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*
no-prop	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*
no-check	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*	*pend.*

Table 2: Main results on MedScope (all 248 examples; offline rows are final). 95% case-level bootstrap intervals are in the appendix. *By construction: the gate never propagates and never writes new content. "—": not applicable (free-text systems give no verdict; the lexical baseline writes no text). Model rows will be filled from `tables/main_medec_scope_v0.tex`.

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A Annotation Decision Procedure

Annotators read all evidence first, then the feedback, and for each claim apply in order: (1) an evidence item contradicts or materially refines the claim, or makes a forward-looking claim moot ⇒

D; evidence items conflict or are inconclusive ⇒ U-evidence; evidence supports it ⇒ go to (4); (2) no evidence speaks to it but it was inferred from a D/P claim and no longer holds ⇒ P; (3) inferred from a U claim ⇒ U-dependency; (4) the feedback disputes it with no evidence either way ⇒ U-feedback, otherwise K. A verbal statement by a senior clinician is feedback, not evidence, unless documented. Generic vague feedback receives the verdict *insufficient* even when evidence settles the claims.

B Example

[TODO: One MedScope case across its conditions (valid / false / authority-only) as a figure.**]**

C Prompts

[TODO: Align, revise, evidence-aware, dialogue, judge prompts verbatim (intexr/scope.py, intexr/prompts.py).**]**